

LEARNING AND PERFORMANCE IN MULTIDISCIPLINARY TEAMS: THE IMPORTANCE OF COLLECTIVE TEAM IDENTIFICATION

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In multidisciplinary teams in the oil and gas industry, we examined expertise diversity's relationship with team learning and team performance under varying levels of collective team identification. In teams with low collective identification, expertise diversity was negatively related to team learning and performance; where team identification was high, those relationships were positive. Results also supported nonlinear relationships between expertise diversity and both team learning and performance. Finally, team learning partially mediated the linear and nonlinear relationships between diversity and performance. Findings broaden understanding of the process by which and the conditions under which expertise diversity may promote team performance.

It is hardly possible to overrate the value . . . of placing human beings in contact with persons dissimilar to themselves, and with modes of thought and action unlike those with which they are familiar. . . . Such communication has always been, and is particularly in the present age, one of the primary sources of progress.

John Stuart Mill,
Principles of Political Economy, 1848

Intimate society between people radically dissimilar to one another is an idle dream. Unlikeness may attract, but it is likeness which retains.

John Stuart Mill,
The Subjection of Women, 1869

In recent years there has been a significant increase in the use of multifunctional or multidisciplinary work teams as a key approach to organizing work (Jackson, 1995; Zakarian & Kusian, 1999). Examples include product development teams, cross-functional teams, brainstorming groups, and management teams. The motivating premise underlying the use of these teams is that when representatives from all of the relevant areas of expertise are

brought together, team decisions and actions are more likely to encompass the full range of perspectives and issues that might affect the success of a collective venture. Multidisciplinary teams are therefore an attractive organizing option when individuals possess different information, knowledge, and expertise that bear on a complex problem or issue.

But although the potential value of multidisciplinary teams remains clear, a growing body of evidence suggests that organizations frequently find it difficult to realize this potential. In fact, the empirical literature examining the performance benefits of expertise diversity in teams has been decidedly equivocal, reporting positive relationships between expertise diversity and performance in some cases and negative or null relationships in other cases (e.g., Bantel & Jackson, 1989; Hambrick, Cho, & Chen, 1996; Murray, 1989; Simons, Pelled, & Smith, 1999). Given this conflicting pattern of results, recent research has begun to question the simplistic diversity-promotes-performance model in order to consider how (via what mediators or intervening variables) and when (in the presence of what moderators) expertise diversity might lead to higher or lower performance. This shift in research focus has led to a number of important insights. For example, scholars now know that expertise diversity is more likely to yield performance benefits in nonroutine task environments (Hambrick et al., 1996; Murray, 1989) and that information sharing–communication is one key mechanism by which expertise diversity might promote performance

A Royal Netherlands Academy of Arts and Sciences Fellowship awarded to Gerben S. Van der Vegt facilitated preparation of this article. The authors would like to thank Marc Anderson, Karen Van Oudenhoven–Van der Zee, Ryan Quinn, and Nico van Yperen for comments and feedback on the manuscript, and three anonymous reviewers for excellent guidance during the review process.

(Bantel & Jackson, 1989; Bunderson & Sutcliffe, 2002).

Nevertheless, academic understanding of the processes by which and the conditions under which expertise diversity promotes or hinders group performance is far from complete, and some fundamental pieces are missing. For example, past research has tended to focus on contextual moderators as explaining the conditions under which expertise diversity will be associated with higher or lower performance (e.g., Jehn, Northcraft, & Neale, 1999; Scott, 1997; Simons et al., 1999). This approach ignores the very real possibility that while the external context may require greater expertise diversity, a diverse team may be unable to utilize diverse member inputs because of tendencies toward “social categorization” and “homophily.” So although the external context surrounding a team suggests conditions under which expertise diversity is *beneficial*, one must consider the motivational climate within a team in order to identify conditions under which expertise diversity will be *leveraged*. We will argue that this motivational climate begins with members’ shared sense of identification with a group and that the effect of expertise diversity on team outcomes is therefore contingent on the degree of “collective team identification” within a group.

Furthermore, one of the principal explanations for how expertise diversity might be beneficial is that a diversity of knowledge and expertise within a group can promote learning and search behaviors that in turn lead to adaptive, innovative solutions. But while the relationship between expertise diversity and innovative solutions does appear to be supported in past research (e.g., Ancona & Caldwell, 1992; Bantel & Jackson, 1989; Tsui, Egan, & Xin, 1995; Wiersema & Bantel, 1992), the intrateam process through which this demography-outcome relationship plays out has not been carefully examined. In order to better understand how expertise diversity might translate into team performance, in the present study we attempted to open this black box (Lawrence, 1997) by explicitly considering the mediating role of team learning behaviors.

Finally, in past research on the relationship between expertise diversity and performance, a monotonic relationship has been implicit; the assumption has been that increased expertise diversity leads to corresponding increases or decreases in some outcome variable. Positing a monotonic relationship ignores the possibility that there may be nonmonotonically decreasing or increasing returns to greater diversity within a group. Another purpose of this study, therefore, was to advance understanding of the relationship between exper-

tise diversity and group performance by considering possible nonmonotonic effects.

CONCEPTUAL BACKGROUND AND HYPOTHESES

The term “expertise diversity” refers to differences in the knowledge and skill domains in which members of a group are specialized as a result of their work experience and education. Although these knowledge and skill domains may correspond to the functional groupings used to partition labor in contemporary organizations (e.g., finance, operations, engineering, R&D), functional groupings are often a crude indicator of the actual task specializations that exist in real work teams. For example, a team of research scientists may be very different in their technical specializations but still be working within the same department or functional area. So the concept of expertise diversity as used here resembles the concept of “functional assignment diversity” as defined by Bunderson & Sutcliffe (2002) but is not fully analogous. Our present concept differs in that it embraces the possibility of more finely graded expertise distinctions than just functional affiliation. Expertise diversity within a team is maximized when members’ areas of expertise reflect equal representation of a relevant set of expertise domains and is minimized when team members all come from the same domain.

We define *collective team identification* as the emotional significance that members of a given group attach to their membership in that group. This definition acknowledges that social identification is multidimensional, encompassing “*knowledge of . . . membership in a social group (or groups) together with the value and emotional significance attached to that membership*” (Tajfel, 1978: 63; emphasis added). Past research has suggested that these three components of social identification—cognitive, evaluative, and emotional—are empirically distinct and relate differently to key outcome variables (see Bergami & Bagozzi, 2000; Ellemers, Kortekaas, & Ouwerkerk, 1999). Of these three components, the emotional component has been shown to most clearly “supply the motivational force” leading to action or the “readiness to engage in or disengage from interaction” (Bergami & Bagozzi, 2000: 563). Since we are interested in collective team identification as a motivational force that can enable interaction in the face of diversity, we focus on the emotional component of social identification in this study.

The construct of collective team identification as conceptualized here is theoretically similar to the construct of team-level affective commitment (e.g.,

Bishop & Scott, 2000; Kirkman & Rosen, 1999), since both constructs focus on the emotional significance attached to team membership. But while the two constructs may be similar at a phenomenological level, they differ in the theoretical lens that is used to illuminate the phenomenon—social identification or group commitment. In this article, we adopt a social identification lens and therefore focus on collective team identification. Furthermore, since collective team identification gets at an individual's emotional relationship with a team rather than his/her relationship with other team members, it differs from constructs like team cohesiveness, which concern the quality of interpersonal relationships among employees (Scott, 1997).

Finally, we define team learning behaviors as activities by which team members seek to acquire, share, refine, or combine task-relevant knowledge through interaction with one another (Argote, Gruenfeld, & Naquin, 1999: 370). These activities may include asking questions, challenging assumptions, seeking different perspectives, evaluating alternatives, and reflecting on past actions (Edmondson, 1999, 2002; Gibson & Vermeulen, 2003). We therefore view team learning behavior as one aspect of a group's "interaction process" (Hackman & Morris, 1975) or as an example of a "group action process" (Marks, Mathieu, & Zaccaro, 2001). Likewise, we assume that group interaction processes are those activities by which "input factors affect performance outcomes" (Hackman & Morris, 1975).

Expertise Diversity, Learning, and Performance in Groups

Scholars have long recognized that exposure to individuals with different expertise, knowledge, and experience is a key source of individual and collective learning. Interaction with dissimilar others promotes learning and innovation by exposing individuals to new paradigms and perspectives and by enabling (and often requiring) the cross-fertilization of ideas. Consistent with this premise, past research in organizational settings has suggested that diverse groups tend to be more creative and innovative. Bantel and Jackson (1989) found that diversity in functional backgrounds was associated with more administrative innovations in a sample of bank management teams. Ancona and Caldwell (1992) found that new product teams whose members were from a more diverse set of functional areas communicated more outside their teams, which led to more creative solutions. Wiersema and Bantel (1992) found that management teams composed of individuals with diverse educational specializations were more likely to engage in

change. And reviews of empirical research on group diversity have concluded that teams produce more creative solutions when they are composed of individuals with diverse sets of backgrounds and experiences (Jackson, 1992; Milliken & Martins, 1996; Tsui et al., 1995).

But the findings have not been wholly consistent. For example, Ancona and Caldwell (1992) found that although diversity in functional assignments was associated with greater external communication, which was in turn associated with greater innovation, the direct effect of functional diversity on innovation was negative. Furthermore, there is no consistent evidence that expertise diversity is associated with higher performance, and some evidence has demonstrated a negative relationship (see the reviews by Jackson [1992], Milliken and Martins [1996], Tsui et al. [1995], Webber and Donahue [2001], and Williams and O'Reilly [1998]). In other words, exposure to a diverse set of backgrounds, experiences, and perspectives within a team may not always promote team innovativeness or team performance and may, in fact, detract from both. This is a puzzling pattern of results.

We propose that in order to solve this puzzle, researchers must first open the black box of team process to see how expertise diversity translates into improved or impaired performance (Lawrence, 1997). Specifically, it needs to be determined whether a given relationship between expertise diversity and adaptive team outcomes does, in fact, occur through integrative learning and the cross-fertilization of ideas, as has been assumed in past research (e.g., Kanter, 1988: 175). If such a relationship can be seen, then the inconsistent findings presented in past research may be the result of the presence or absence of team learning behaviors. It would then follow that a careful and explicit consideration of the conditions under which team learning behaviors will be present or absent can point to conditions under which expertise diversity will result in improved or impaired performance.

A second key to understanding the above pattern of results is to recognize that a learning-oriented response to expertise diversity can be threatening and uncomfortable. As individuals specialize in different functions, departments, and disciplines, they not only gain specialized knowledge and skill related to their chosen area, but also develop personal identities that are linked to that area of specialization (Scott, 1997). That is, areas of specialization within an organization represent salient social categories that individuals use to think about themselves and others. Theory and research have suggested that individuals are motivated to interact most with members of their own social categories

(McPherson, Smith-Lovin, & Cook, 2001) and to emphasize the positive aspects of their categories in relation to other categories (which end up suffering by comparison) (see Tajfel & Turner, 1979). In other words, there are well-documented reasons why one would expect that members of a multidisciplinary or multifunctional team might find it difficult to engage the other members of their team in critical and investigative interaction processes (Amason, 1996). In fact, the natural tendency may be to stereotype the other members of one's team, to assume that they "just don't understand," and to argue and defend rather than seek conciliation and integration.

Collective Team Identification as a Moderator

Members of a multidisciplinary team need a motivation to transcend these disruptive tendencies before they will exert the effort to do so. We argue that this motivational element begins with collective team identification as defined above—that is, with a sense that membership in one's team is an emotionally significant aspect of one's identity. In teams with high levels of collective team identification, individuals are committed to the team and its goals rather than (or in addition to) their own goals or the goals of their particular specialty areas. This broadening of team members' relevant units of affiliation is a critical prerequisite to the effective integration of divergent ideas and perspectives.

Few diversity researchers have incorporated the concept of collective team identification into their models. In diversity studies that have examined the role of team identity (e.g., Earley & Mosakowski, 2000) or the related individual-level concept of team identification (e.g., Chattopadhyay, 1999; Scott, 1997; Van der Vegt, Van de Vliert, & Oosterhof, 2003), researchers have generally assumed that these variables mediate the relationship between demographic diversity and a number of outcome variables. That is, they have assumed that individuals will find it more difficult to identify with a diverse team and that this lack of identification will have important implications for various team outcomes.

Results from past research do seem to support the notion that demographic diversity complicates identification with a team, yet the evidence also suggests that the relationship between diversity and identification can vary across teams. In fact, because a variety of factors other than demographic diversity can influence team identification (e.g., Bergami & Bagozzi, 2000; Ellemers et al., 1999), we would expect that diversity and identification will be only loosely coupled in teams. That is, demo-

graphically diverse teams can be high on team identification, while demographically homogeneous teams can be low on team identification. And if this is the case, it seems quite possible that the effects of diversity on various team outcomes will look different in high-identification teams than in low-identification teams. Specifically, one might expect that diverse teams will be better able to exchange information and learning across specialization boundaries when there is a shared sense of team identification than when there is not.

This view is consistent with self-categorization theory, in which it is recognized that intergroup comparisons often occur within the context of higher-order similarity (e.g., Hornsey & Hogg, 2002; Turner, 1975) and that recategorization can mitigate the potentially adverse effect of subgroup identities (Brewer & Miller, 1984). A series of experimental and survey studies testing the common in-group identity model of intergroup relations (Gaertner, Dovidio, & Bachman, 1996), for example, has shown that inducing an inclusive superordinate identity in the minds of the members of two separate subgroups reduces intergroup bias and stereotyping. A strong superordinate identity makes it possible for people to categorize dissimilar persons as in-group members rather than as out-group members. Consequently, evaluations of those persons become more positive (Tajfel & Turner, 1979), group-based biases in perception, recall, and attribution decrease (Hewstone, 1990), and different subgroups are cast in a complementary role relation (Hornsey & Hogg, 2002). Taken together, this evidence suggests that for members of a team working together with a high level of collective team identification, expertise diversity should stimulate team learning and enhance adaptive performance.

This evidence also suggests that multidisciplinary teams with low levels of collective team identification should be characterized by a retention of specialty area identities, biases, and stereotypes, which should increase members' tendency to overlook or reject the information and perspectives of members from other specialty areas (Scott, 1997). As a consequence, expertise diversity is likely to be negatively related to team learning behavior and performance in teams with low collective team identification. Consequently, we propose:

Hypothesis 1. Collective team identification moderates the relationship between expertise diversity and team learning behaviors; the relationship is negative when collective team identification is low but positive when collective team identification is high.

Hypothesis 2. Collective team identification moderates the relationship between expertise diversity and team performance; the relationship is negative when collective team identification is low but positive when collective team identification is high.

The Mediating Role of Team Learning Behavior

Previous research has established that process variables like intrateam conflict (e.g., Jehn et al., 1999; Simons et al., 1999), and external communication (e.g., Ancona & Caldwell, 1992) can mediate the relationship between diversity and team effectiveness. In this study, we focused on the mediating role of team learning behavior. We argue that the interaction between expertise diversity and collective team identification will be positively associated with team learning behavior, and that team learning behavior, in turn, will enhance overall team effectiveness by promoting continuous process improvement. This latter effect is consistent with empirical evidence suggesting that when team members engage in learning-related behaviors, a team improves its ability to adapt to its environment in order to operate effectively within that environment (e.g., Edmondson, 1999).¹ Thus, we suggest that the interactive effect of expertise diversity and collective team identification on performance occurs, at least in part, through team learning.

Hypothesis 3. Team learning behavior mediates the relationship between expertise diversity and team performance.

Nonlinear Effects of Expertise Diversity

Although most diversity studies have examined linear effects of team diversity, recent empirical research suggests that the relationship between diversity and a number of outcome variables may in fact be nonlinear (e.g., Earley & Mosakowski, 2000; Gibson & Vermeulen, 2003). We suggest here that similar nonlinear effects may occur for expertise diversity, and the above hypotheses should therefore be amended to accommodate a nonlinear effect.

We argued earlier that expertise diversity would be positively related to team learning and performance in teams with high levels of collective team

identification, since a shared sense of identification with a team allows team members to transcend specialization differences in order to learn from and leverage one another's differences. Beyond some optimal level of expertise diversity, however, the different perspectives and opinions resulting from high levels of expertise diversity may lead to information overload and increase the general complexity of team problem solving (Ancona & Caldwell, 1992; Milliken & Martins, 1996). As a result, while members may still be motivated to integrate diverse domains of expertise in a team with very high levels of expertise diversity, they may be unable to do so, given complex tasks. This line of argument suggests that the relationship between expertise diversity and both team learning and team performance in teams with a high collective team identification will be positive up to a point and will then begin to decline, resulting in an inverted U-shaped pattern.

We also argued above that in teams with low levels of collective team identification, the relationship between expertise diversity and both team learning and team performance will be negative as increasing diversity in member backgrounds and experiences tends to increase social categorization and in-group biasing. However, beyond some moderate level of expertise diversity, the tendency to categorize and stereotype other team members should decrease, because fewer bases for subgroup formation, categorization, and social identity exist given that "everybody is different" (Gibson & Vermeulen, 2003: 209; see also Earley & Mosakowski, 2000). As a consequence, beyond some moderate level of expertise diversity, we would expect no further decreases in team learning and performance and, instead, would expect to see a positive effect of diversity on learning and performance. Any increase would be mitigated, however, by the fact that very high levels of expertise diversity will still be accompanied by information overload (as argued above) and decision-making difficulty. The result of this dynamic would be a U-shaped relationship between expertise diversity and both team learning and team performance in teams with low levels of collective team identification. Taken together, these arguments suggest the following extensions to the arguments formalized in Hypotheses 1 and 2:

Hypothesis 4. Collective team identification moderates a curvilinear relationship between expertise diversity and team learning behavior: There is a U-shaped relationship when collective team identification is low and an inverted

¹ Note that our team performance focus here is on overall team effectiveness since past research has demonstrated that a strong learning emphasis in teams can compromise short-term efficiency (Bunderson & Sutcliffe, 2003).

U-shaped relationship when collective team identification is high.

Hypothesis 5. Collective team identification moderates a curvilinear relationship between expertise diversity and team performance: There is a U-shaped relationship when collective team identification is low and an inverted U-shaped relationship when collective team identification is high.

Finally, for the reasons noted above, we would expect that team learning behavior will mediate these nonlinear relationships between expertise diversity and team performance. The following extension of Hypothesis 3 is therefore warranted:

Hypothesis 6. Team learning behavior mediates the moderated, nonlinear relationship between expertise diversity and team performance.

METHODS

Sample and Data Collection

We tested the above hypotheses using data obtained from multidisciplinary teams working within a "Global 1000" (*BusinessWeek*, 2003) company in the oil and gas industry in the Netherlands. These teams are composed primarily of scientists, engineers, and technicians and are responsible for research and development functions such as identifying new natural gas and petroleum locations, implementing new technologies, collecting and analyzing field data, and planning short- and long-term projects. The company formed these teams for one of the above purposes by assembling individuals with ranges of relevant expertise. Each employee was permanently assigned to just one ongoing team. In order to perform their work, team members were required to interact frequently, share resources and information, and coordinate efforts toward the accomplishment of joint goals. Although each team received regular team-level performance feedback, there were no formal individual or group incentive plans in this organization. Discussions and observations clearly suggested that these groups functioned as teams and that they were seen by themselves and others as teams.

With the authorization of top management, we visited the supervisors of all 62 teams in this organization to ask for their participation, promising feedback should they choose to participate. In order to balance our data requirements with a call from management to minimize time demands, we used an "informant sampling approach" (Van de Ven & Ferry, 1980). An infor-

mant sampling approach recognizes that many members of a given collective are qualified to provide assessments of those global properties that they experience together. The informant approach therefore "relies on a limited selective sample of people who are the most knowledgeable of the global properties of interest" (Van de Ven & Ferry, 1980: 72) rather than seeking to obtain measures from all members of a collective. Furthermore, since some variance across informants is to be expected, the informant approach involves sampling several informants so that interrater reliability can be assessed and, if convergence is demonstrated, a "balanced perspective" can be obtained by "averaging informants' perceptions" (see Van de Ven & Ferry, 1980: 73). Simons and colleagues (1999) and Tsui, Egan, and O'Reilly (1992) provided examples of using an informant sampling strategy to study team diversity.

In accordance with this approach, we asked supervisors from those teams who indicated their willingness to participate to complete and return a supervisor survey measuring team performance. We also requested that each supervisor ask four members of her/his team to complete and return a team member survey measuring collective team identification and the team process variables. We received 58 supervisor questionnaires (94%) and 225 team member questionnaires (91%). In addition to the supervisor and subordinate survey data, we also collected personnel data on each of the teams in the sample from the human resources department (e.g., employee demographics, team size, etc.).

Our final sample consisted of those 57 teams from which we had received three or more subordinate surveys and a supervisor survey, and for which archival data were available. These teams ranged in size from 4 to 26 members ($\bar{x} = 11.7$, $s.d. = 6.1$); 84 percent of team members were male; the average age was 42.7 years ($s.d. = 7.7$), and the average time with the current team was 1.5 years ($s.d. = .5$). In addition, 59 percent of team members held master's degrees or higher, and an additional 13 percent held bachelor's degrees.

Measures

Following our informant sampling approach (described above), we framed all the items on the supervisor and team member surveys as "informant" rather than "respondent" items (see Van de Ven & Ferry, 1980); informant items ask individuals to evaluate their team rather than their own personal behaviors or attitudes. Unless otherwise

stated, we assessed all questionnaire items using a scale ranging from 1, "completely disagree," to 7 "completely agree." All items were drawn from published scales. Items were discussed with two company representatives before the questionnaires were distributed. This process resulted in our making several changes to the wording of items in order to increase clarity and interpretability.

Dependent and independent variables. *Expertise diversity* was computed from archival data. Seven broad disciplinary areas were represented among the members of these teams: geoscience, petroleum engineering, field engineering, production engineering, well engineering, information and communication technology, and administration; other areas were coded into an "other" category. Furthermore, each category included a number of specializations. Geoscience, for example, consisted of 12 different specializations, such as regional evaluation (basin and plan evaluation), production geology (field evaluation and static reservoir modeling), seismic data acquisition, and "geoinformation" management. In total, the archival data revealed 50 specializations. Discussions with two experts from the human resources department suggested that these specializations were highly relevant and captured key sources of expertise diversity in these teams. We therefore computed expertise diversity at the specialization level using Blau's (1977) formula, $1 - \sum p^2$, where p is the proportion of a group in the i th category. A higher index score indicated greater expertise diversity among team members.

Collective team identification was computed using items from Allen and Meyer's (1990) affective commitment scale. Ellemers et al. (1999; see also Bergami & Bagozzi, 2000) suggested that the construct of affective commitment captures the emotional component of social identification because affective commitment concerns "identification with, involvement in, and emotional attachment to the [collective]" (Allen & Meyer, 1996: 253). We therefore measured team identification using the four highest-loading items from Allen and Meyer's (1990) affective commitment scale (as did Bergami and Bagozzi [2000]). Specifically, we asked team member informants to assess the extent to which members of their team "feel emotionally attached to their team," "feel a strong sense of belonging to their team," "feel as if the team's problems are their own," and "feel like part of the family in their team." Cronbach's alpha for this scale was .92.

Team learning behavior was measured using four items adapted from Edmondson (1999) and Drach-Zahavy and Somech (2001). These items ask team informants to evaluate the extent to which mem-

bers of their team "criticize each other's work in order to improve performance," "freely challenge the assumptions underlying each other's ideas and perspectives," "engage in evaluating their weak points in attaining effectiveness," and "utilize different opinions for the sake of obtaining optimal outcomes" ($\alpha = .75$).

Team performance was measured using supervisor ratings of three performance criteria based on previous research (Ancona & Caldwell, 1992) and two additional criteria suggested by managers in the host company. The three established criteria were efficiency, quality, and overall achievement; the two other criteria were productivity and mission fulfillment. The response set for these items ranged from 1, "far below average," to 7, "far above average." Each supervisor was asked to compare the performance of his or her team with the performance of teams that performed similar tasks. Cronbach's alpha for this scale was .87.

Control variables. Given that groups varied considerably in size and that prior research has demonstrated a relationship between team size and both cohesiveness and internal communication, we included *team size* as a control variable in all analyses (Ancona & Caldwell, 1992; Bantel & Jackson, 1989). *Team tenure diversity*, *age diversity*, *gender diversity*, and *nationality diversity* were also included as controls since prior work has suggested that these variables are related to interpersonal contacts, knowledge bases, and performance (Ancona & Caldwell, 1992). The coefficient of variation (standard deviation divided by the mean) was used for the continuous demographic variables of team tenure and age, and the Blau (1977) index was used for gender and nationality.

In order to establish the robustness of the mediating effect of team learning behavior, we controlled for the effect of three other variables that have been shown to mediate the relationship between team diversity and performance outcomes in past diversity research: task and relationship conflict (Jehn et al., 1999), and external communication (Ancona & Caldwell, 1992; Keller, 2001; Scott, 1997). We measured task conflict using the following three items ($\alpha = .83$) adapted from Jehn et al. (1999): "How much conflict of ideas is there in this team?", "How often do the members of this team disagree about work things?", and "How different are members' viewpoints on decisions?" We also measured relationship conflict using three items ($\alpha = .71$) adapted from Jehn et al. (1999): "How much are personality clashes evident within this team?", "How much do you think that members do not interpersonally get along?", and "How often do the members of this team disagree about non-work

(social or personality) things?" (1 = "not much"; 7 = "very much"). External communication was measured with three items ($\alpha = .88$) adapted from Ancona and Caldwell (1992) and Keller (2001): "The members of this team frequently talk to members of other teams," "The members of this team feel comfortable contacting members from other teams when the need arises," and "Team members go out and get all the information they possibly can from other teams in this organization."

Confirmatory factor analysis. Using confirmatory factor analysis, we assessed the convergent and discriminant validity of all of the survey scales: collective team identification, team learning, task and relationship conflict, and external communication. We computed parameter estimates with the LISREL 8 computer package, using the maximum likelihood method. The expected five-factor model did not yield an adequate overall fit to the data ($\chi^2[109, 225] = 394.48, p < .001$), given that there were high cross-loadings between task and relationship conflict items. We therefore examined the fit of a four-factor model in which the task and relationship conflict items all loaded on one factor. This model resulted in a much better fit of the measurement model to the data ($\Delta\chi^2[4] = 113.61, p < .001$). The standardized root mean square of the residuals was .06; the goodness-of-fit index was .89; and the comparative fit index was .91. The factor "loading" for each item on its corresponding construct was significant at the .001 level or better. On the basis of these results, we decided to combine the task and relationship conflict items into one overall *intrateam conflict* scale ($\alpha = .89$).²

A separate confirmatory factor analysis on the supervisor ratings of team performance revealed a good fit of the model to the data ($\chi^2[5, 58] = 2.04, n.s.$). The standardized root mean square of residuals was .02, and the goodness-of-fit index and the comparative fit index both exceeded .90.

Interrater agreement and reliability. As noted above, we sampled several informants from each team in assessing collective team identification, team learning, intrateam conflict, and external communication, under the assumption that informant ratings would reflect a shared reality within each team. If this assumption is valid, we would

expect to find that ratings from different informants on the same team are similar to one another and, furthermore, that they are more similar to one another than they are to informant ratings from other teams (see Bliese, 2000). We examined this expectation using the average interrater agreement coefficient (r_{wg} ; James, Demaree, & Wolf, 1984) and the intraclass correlation coefficient (ICC[1] and ICC[2]; Bliese, 2000). Median r_{wg} values were .95 for collective team identification, .93 for team learning, .84 for intrateam conflict, and .88 for external communication, suggesting that informant ratings within a given team were highly consistent with one another.

One-way analyses of variance suggested that informant ratings of collective team identification, team learning, intrateam conflict, and external communication all differed significantly ($p < .001$) between teams. Furthermore, ICC(1) values easily cleared established "hurdle rates" (see James, 1982) with values of .40 for collective team identity, .35 for team learning, .30 for intrateam conflict, and .26 for external communication, suggesting that much of the variance in ratings is due to team membership (Bliese, 2000). Finally, the reliability of the group means, as measured by the ICC(2) coefficient, was .71 for collective team identification, .66 for team learning, .60 for intrateam conflict, and .56 for external communication.

Together, the above analyses supported the aggregation of informant responses from a team to create team-level variables for collective team identification, team learning, intrateam conflict, and external communication.

Analysis

We conducted hierarchical multiple regression analyses to test our hypotheses (Aiken & West, 1991; Cohen, Cohen, West, & Aiken, 2003). We examined the following equations in four steps in order to isolate the contribution of different terms:

$$Y = \mathbf{B}_1\mathbf{C} + b_2X + b_3Z + b_4XZ + b_5X^2 \quad (1)$$

$$+ b_6X^2Z + e.$$

$$M = \mathbf{B}_1\mathbf{C} + b_2X + b_3Z + b_4XZ + b_5X^2 \quad (2)$$

$$+ b_6X^2Z + e.$$

where Y was performance, M was team learning, $\mathbf{C}$ was the vector of control variables (with $\mathbf{B}_1$ a corresponding vector of beta coefficients), X was expertise diversity, and Z was collective team identification. Control variables were entered alone in

² We also tested the fit of a model in which team learning and external communication loaded on the same factor given the high correlation between these two constructs (see Table 1). This three-factor model resulted in a significantly worse fit ($\Delta\chi^2[3] = 110.23, p < .001$) than the four-factor model, supporting the discriminant validity of the learning and external communication scales.

step 1, “main effects” (X and Z) were entered in step 2, the linear-by-linear interaction term (XZ) was entered in step 3, and the quadratic (X^2) and linear-by-quadratic (X^2Z) interaction terms were entered in step 4. Independent variables were standardized to facilitate interpretation.

To test for the mediating effect of team learning, we followed the procedures outlined by Baron and Kenny (1986). This procedure involves adding a third equation to the two equations described above:

$$Y = \mathbf{B1C} + b_2X + b_3Z + b_4XZ + b_5X^2 + b_6X^2Z + M + e. \quad (3)$$

Mediation is supported if main effects and interaction terms are significant in Equations 1 and 2 but not in Equation 3.

RESULTS

Table 1 presents means, standard deviations, and zero-order Pearson correlations among study variables. The results of this univariate analysis showed that expertise diversity was not linearly related to the measures of collective team identification, team learning, and team performance. However, as expected, team learning was positively associated with supervisor-ratings of team performance ($r = .38, p < .01$).

Table 2 presents the results of the hierarchical regression analyses. All regressions met the major model assumptions; that is, no serious violations were found in the plots of standardized residuals as compared to the predicted values, in the normal probability plots of standardized residuals, and with regard to the independence of error terms (the

Durbin-Watson statistic ranged from 1.73 for team performance to 1.83 for team learning behavior).

Hypotheses 1 and 2 state that collective team identification moderates the relationship between expertise diversity and both team learning and team performance. As shown in Table 2, after entering the control variables (step 1) and the main effects of expertise diversity and collective team identification (step 2), the interaction of expertise diversity and collective team identification reached significance for team learning ($\Delta R^2 = .11, p < .01$; see step 3 of model 1) and team performance ($\Delta R^2 = .18, p < .01$; see step 3 of model 2). Simple slope tests, following the procedure outlined by Aiken and West (1991), revealed that for teams with low levels of collective team identification, expertise diversity was negatively related to both team learning ($b = -.32, t = -3.07, p < .01$) and team performance ($b = -.28, t = -2.44, p < .05$). For teams with a high level of collective team identification, expertise diversity was positively related to team learning ($b = .20, t = 1.78, p = .08$) and team performance ($b = .28, t = 2.10, p < .05$). These results are consistent with Hypotheses 1 and 2.

Hypothesis 3 states that team learning mediates the interactive effect of expertise diversity and collective team identification on team performance. A comparison of the significant interaction effect in model 2 with the interaction effect in model 3 shows that the addition of team learning as a control variable in step 1 of model 3 diminished the effect on team performance of the interaction of expertise diversity and collective team identification ($\Delta R^2 = -.11$). Although the interaction term remained significant, the results of a Sobel test (MacKinnon, Warsi, & Dwyer, 1995) showed that the mediated effect was significant ($Z = 2.15, p <$

TABLE 1
Means, Standard Deviations, and Correlations^a

Variable	Mean	s.d.	1	2	3	4	5	6	7	8	9	10
1. Team size	11.71	6.10										
2. Age diversity	0.16	0.05	.22									
3. Team tenure diversity	0.35	0.17	-.14	.38**								
4. Gender diversity	0.26	0.25	.04	.17	.02							
5. Nationality diversity	0.23	0.22	.03	.46**	.29*	-.05						
6. Expertise diversity	0.52	0.30	.06	.26*	.14	-.07	.47**					
7. Collective team identification	4.53	0.85	-.03	-.21	-.05	.14	-.18	-.12				
8. Team learning	4.60	0.75	-.19	.01	.16	.17	.17	-.09	.32*			
9. External communication	5.25	0.87	-.18	.16	.32*	.24	.25	.04	-.02	.66**		
10. Intrateam conflict	2.74	0.81	.17	.14	-.03	.10	.18	.09	-.47**	.43**	-.07	
11. Team performance	5.31	0.70	.01	.05	.13	-.13	.02	.01	.10	.38**	.22	.03

^a $n = 57$.

* $p < .05$

** $p < .01$

TABLE 2
Results of Regression Analysis^a

Independent Variables	Model 1: Team Learning		Model 2: Team Performance		Model 3: Team Performance		Model 4: Team Performance	
	Entry <i>b</i>	Final <i>b</i>	Entry <i>b</i>	Final <i>b</i>	Entry <i>b</i>	Final <i>b</i>	Entry <i>b</i>	Final <i>b</i>
Step 1: Control variables								
Team size	-.09	-.11	.10	.02	.10	.01	.10	.00
Age diversity	-.15	-.07	.05	.05	.05	.01	.05	.04
Team tenure diversity	.11	.06	.15	.06	.15	.09	.15	.09
Gender diversity	.12	.14	-.32*	-.18	-.32*	-.15	-.32*	-.24*
Nationality diversity	.14	.26*	-.19	-.14	-.19	-.10	-.19	-.18
Team learning behavior					.29**	.21*	.36**	.28*
External communication							.11	.10
Intrateam conflict							.29*	.28*
<i>F</i>	1.21		1.37		2.27		2.16	
<i>R</i> ²	.11		.09		.27		.39	
Step 2: Main effects								
Expertise diversity	-.11	-.18	-.01	-.18	.11	.01	.14	.02
Collective team identification	.22*	.54**	.10	.42**	-.09	.13	.04	.17
ΔR^2	.03		.01		.03		.03	
<i>F</i>	2.14		1.48		1.66		1.41	
<i>R</i> ²	.14		.10		.30		.42	
Step 3: Interaction								
Expertise diversity $\times$ collective team identification	.21**	.01	.31**	.10	.19*	.11	.13	.06
ΔR^2	.11		.18		.07		.04	
<i>F</i>	3.03		2.14		2.53		2.08	
<i>R</i> ²	.25		.28		.37		.46	
Step 4: Quadratic effects								
Expertise diversity squared		-.02		-.08		-.06		-.09
Expertise diversity squared $\times$ collective team identification		-.28**		-.27*		-.13		-.09
ΔR^2	.08		.07		.02		.01	
<i>F</i>	3.42		2.58		-2.37		2.07	
<i>R</i> ²	.33		.35		.39		.47	

^a $n = 57$. Unstandardized regression coefficients and unadjusted R^2 values are reported. The final b 's represent the unstandardized regression effects from the last step of the analyses.

* $p < .05$

** $p < .01$

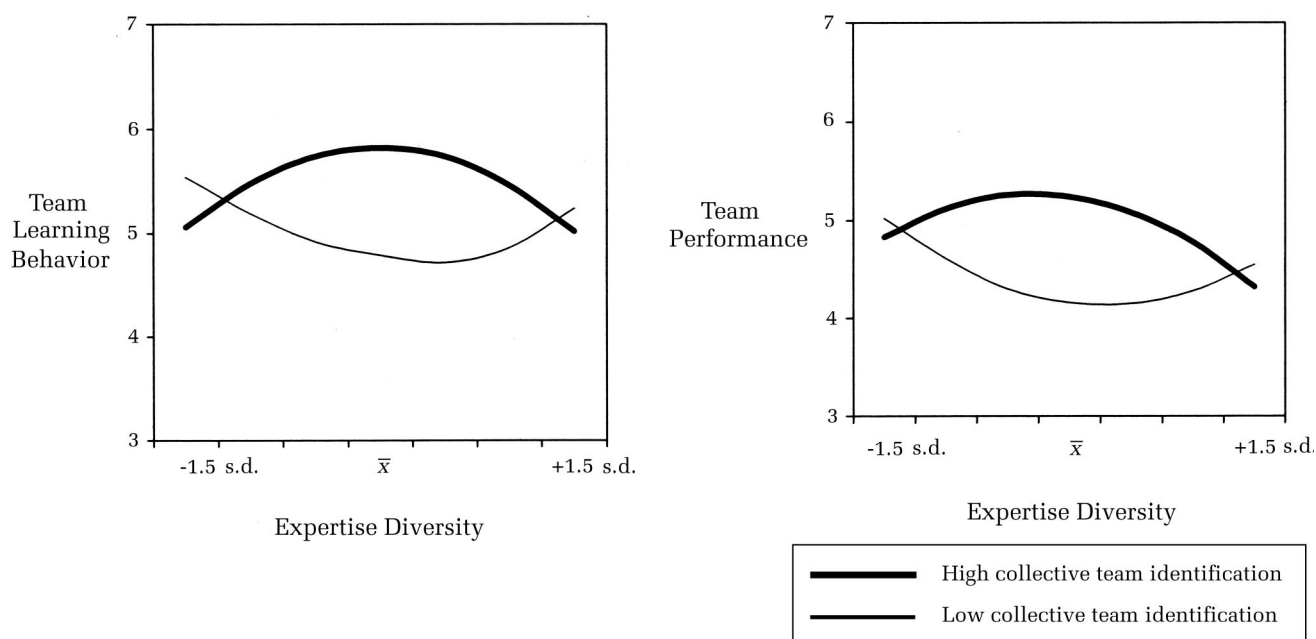
.05). These results imply that team learning *partially* mediates the effects of the interaction between expertise diversity and collective team identification on team performance.

Hypotheses 4 and 5 posit a nonlinear relationship between expertise diversity and both team learning and team performance, moderated by the degree of collective team identification. We tested these hypotheses by entering the quadratic and quadratic-by-linear interaction of expertise diversity and collective team identification into the regression equation in step 4. The hypothesized multiplicative interaction of expertise diversity

squared by collective team identification was significant for team learning ($\Delta R^2 = .08$, $p < .01$) and team performance ($\Delta R^2 = .07$, $p < .05$). Note also that when the quadratic effects are added to the regression equation, the expertise diversity-collective team identification interaction term drops from significance, suggesting that the moderated relationship between expertise diversity and both team learning and team performance is more completely described using nonlinear terms.

To facilitate interpretation of this effect, Figure 1 illustrates the relationships between expertise diversity and both team learning and team perfor-

FIGURE 1
Relationships between Expertise Diversity, Team Learning, and Team Performance



mance (see Aiken and West [1991: 12–14] for exact procedures). Figure 1 shows that the relationship between expertise diversity and team learning follows a U-shaped pattern in teams with low levels of collective team identification and an inverted U-shaped pattern in teams with high levels of collective team identification. For team performance, the pattern of results was almost identical. These findings support the nonlinear relationships formulated in Hypotheses 4 and 5.

Hypothesis 6 states that team learning mediates the quadratic-by-linear effect of expertise diversity and collective team identification on team performance. Regression results testing this hypothesis appear in models 2 and 3 of Table 2. Specifically, whereas the quadratic-by-linear effect of expertise diversity and team identification on performance was significant in model 2 ($p < .05$), it dropped from significance when team learning was added in model 3 (step 1). Additionally, the results of a Sobel test suggested that this mediated effect was significant ($Z = 2.08$, $p < .05$). This pattern of results is consistent with Hypothesis 6.

In order to examine whether the mediating effect of team learning behavior observed here is separate from the effect of other variables shown to mediate the diversity-performance relationship in past research, we added intrateam conflict and external communication to model 3. The results of this analysis (presented as model 4 in Table 2) indicated that while intrateam conflict was also significantly

and positively related to team performance, the effect of team learning behavior remained significant. The relationship between external communication and team performance was not significant. In subsequent analyses (not reported in Table 2), we regressed intrateam conflict and external communication on the set of independent variables in model 2. Expertise diversity, alone or in interaction with collective team identification, was not a significant predictor of either variable. These results support the independent effect of team learning behavior in mediating the relationship between expertise diversity and performance in these groups and suggest that neither external communication nor intrateam conflict played a significant mediating role.

DISCUSSION

Toward a More Nuanced Theory of Expertise Diversity and Group Performance

The results of this research have important implications for the way scholars think about the benefits and challenges of expertise diversity in groups. Specifically, these results further underscore the need to move beyond the simple diversity-affects-performance model in order to think in more complex ways about *how* and *under what conditions* a diversity of expertise in groups might promote or inhibit group effectiveness. For exam-

ple, this study suggests that in order to understand whether a given level of expertise diversity in a group will have positive or negative implications for group performance, researchers need to consider the motivational climate that exists within the group and, more specifically, the extent to which members feel emotionally identified with their group. In this sample of teams, collective team identification had a significant moderating effect on the relationship between expertise diversity and group performance: a moderate level of expertise diversity could be associated with either the highest or lowest levels of performance, depending on whether and to what extent members identified with their teams. This finding extends past research on the conditions under which expertise diversity might facilitate or hinder group performance by pointing to the significant role of intragroup factors.

Furthermore, the results of this research clearly suggest that this moderated relationship between expertise diversity and performance is nonmonotonic. Specifically, we found that under conditions of low collective team identification, the relationship between expertise diversity and performance was U-shaped, whereas under conditions of high collective team identification, the relationship between expertise diversity and performance followed an inverted U-shaped pattern. In other words, independent of the degree of collective team identification, very low and very high levels of expertise diversity were associated with moderate levels of performance in these groups. Expertise diversity was most strongly associated with team performance at moderate levels of diversity, with the direction of these effects contingent on the degree of collective team identification. These results suggest that our theories and models of the performance implications of expertise diversity in groups must move beyond linear assumptions in order to accommodate nonmonotonic effects.

Finally, the results of this research help to better explain *how* expertise diversity promotes or inhibits performance outcomes by pointing to the important mediating role of team learning behaviors. In this study, team learning behaviors at least partially mediated all of the effects summarized above. Furthermore, the mediating role of team learning behaviors in this study remained significant even after we controlled for two variables that have been shown to mediate the relationship between expertise diversity and performance in past research: external communication (Ancona & Caldwell, 1992) and intrateam conflict (Jehn et al., 1999). These results suggest that one key means by which a diversity of expertise can affect performance out-

comes is by stimulating search and learning behaviors within a team. The findings presented here therefore contribute important empirical evidence to support the claim that, under the right conditions, expertise diversity can be a key activator of intrateam learning and thereby promote overall team effectiveness (see Kanter, 1988).

Team Learning Behaviors versus Conflict and Communication

Past research has suggested that one means whereby expertise diversity might enhance group performance is by facilitating external communication with a richer "database" of diverse external contacts (e.g., Ancona & Caldwell, 1992; Keller, 2001; Scott, 1997). The present study did not support this expectation; external communication was weakly related to both expertise diversity and performance in these teams. Furthermore, the relationship between team learning behavior and performance remained significant when external communication was included in the model. These results suggest that, for these teams, the potential benefits of expertise diversity were realized through the cross-fertilization of ideas that occurs through intrateam learning efforts and not through the expansion of information and insight that occurs through extrateam contacts and interactions. The high correlation between team learning and external communication does seem to suggest, however, that going outside a team for information and advice may be one important way in which team learning is manifested.

Our results also fail to support a role for intrateam conflict in mediating the relationship between expertise diversity and team performance. Although intrateam conflict was associated with team performance in a model that included all study variables, the zero-order correlation between team conflict and performance was not significant (suggesting a suppression effect). Furthermore, there was no observed relationship between expertise diversity and intrateam conflict in these groups. Results also suggested that team learning behavior remained a significant predictor of performance when intrateam conflict was included in the regression model. This pattern of results suggests that the performance implications of expertise diversity in this sample of teams cannot be attributed to the positive or negative consequences of conflict between experts, as some past research has suggested (e.g., Amason, 1996; Jehn et al., 1999). We did find, however, that the correlation between team learning behavior and intrateam conflict was positive and significant, suggesting that teams more

actively engaged in learning behaviors are more likely to experience intrateam conflict. So while conflict did not mediate the diversity-to-performance relationship in this study, conflict did appear as part of the larger dynamic associated with expertise diversity in teams.

Social Category Diversity and Collective Team Identification

The focus of this study was on diversity in task-related expertise rather than on diversity in the various social category characteristics of team members (e.g., age, gender, ethnicity, etc.). Past research has suggested that whereas diversity in task-related characteristics can have positive implications for group effectiveness, diversity in social category characteristics can have negative implications (Jehn et al., 1999; Pelled, Eisenhardt, & Xin, 1999). But past research on social category diversity has not examined the moderating effect of collective team identification. It is possible that in teams with high levels of collective team identification, social category diversity might also be positively associated with team learning behavior and team effectiveness, since these social category differences between group members can also stimulate adaptive learning behaviors when members identify with their teams.

In order to test this possibility, we created a measure of social category diversity that combined age diversity, gender diversity, and nationality diversity into a single measure of social category diversity for each team. We then examined whether this diversity measure predicted team learning behavior in interaction with collective team identification. Results revealed a pattern of results similar to that observed for expertise diversity. That is, the linear and linear-by-curvilinear interactions between social category diversity and collective team identification were significant predictors of both team learning and team performance. These findings suggest that, contrary to some past research, social category diversity can be associated with team effectiveness—but only when members identify with their teams.

Limitations

It is important to note certain limitations of this study. First, like much of the existing research in this area (see Ancona & Caldwell, 1992; Edmondson, 1999; Keller, 2001), we measured team performance by means of supervisor ratings. Since we do not have data to show that this perceptual measure of team performance is a predictor of more “objec-

tive” team performance, it is possible that supervisor ratings of performance were somehow biased. Research using more objective team performance measures would provide greater confidence in the robustness of these observed effects.

Second, it is always important to consider alternative causality explanations when one adopts a cross-sectional design. For example, one might argue that the members of a team that received high supervisor ratings would know that they are performing well and, as a result, might become more bullish in their team assessments of other variables (e.g., team learning, collective team identification, conflict, etc.). The fact that team performance was not significantly correlated with collective team identification, perceptions of intrateam conflict, or assessments of external communication in this study suggests that this may not have been the case here. Nevertheless, future research adopting a longitudinal design can help to mitigate any residual concerns.

Third, we examined expertise diversity and team learning across a number of different multidisciplinary teams in only one organization. Testing hypotheses in one organization reinforces confidence in the likelihood that organization-level factors (e.g., organizational culture) did not affect our findings, but it also reduces the generalizability of these results. We don’t know whether we would observe a similar pattern of results in different organizations or with different types of teams. It is possible, for example, that team learning behaviors were a more critical mediator of the relationship between expertise diversity and performance for these R&D teams than they would be for teams engaged in more routine tasks. Or it may be that the “optimum” level of diversity in a high-identification team would be observed at lower levels of expertise diversity in teams that perform more routine tasks. Furthermore, since average tenure in these teams was somewhat low ($\bar{x} = 1.5$ years), we can’t be certain that more established teams would exhibit the same dynamics surrounding expertise diversity. Research examining these issues in longer-tenured teams, in teams performing different tasks, or in teams from other organizations and industries would help to both confirm and extend the theory and hypotheses presented in this paper.

Practical Implications

The present study offers several implications for practitioners trying to manage expertise diversity in multidisciplinary work teams. First, our findings suggest that it is important for managers to create the proper mix of expertise in assembling teams.

Too little or too much expertise diversity within a team may dampen team learning behavior and decrease team performance. Moderate levels of expertise diversity within a team make it more likely that members will utilize their different perspectives and learn from one another—if they also identify with their teams. Consequently, it is important that managers take measures to foster a high level of collective team identification within their moderately diverse teams. Companies can encourage collective team identification by creating the right mix of task and goal interdependence among team members (see Van der Vegt et al., 2003), by showing supporting and recognizing the team, by allowing teams to develop a shared history together rather than changing membership frequently, and by increasing contact among team members (e.g., Scott, 1997).

Conclusion

We began this article with two quotes from John Stuart Mill. In the first quote, Mill suggested that diversity in expertise and knowledge was a key aspect of learning and progress. In the second quote, Mill pointed to the nontrivial (perhaps even insuperable) challenges of encouraging meaningful learning and interaction among dissimilar individuals. The present research suggests that both of these statements are valid and that both statements help to explain why past research on the performance benefits of expertise diversity in multidisciplinary teams has been equivocal. Furthermore, the present examination of expertise diversity in a sample of multidisciplinary teams in a Global 1000 organization points to the critical importance of collective team identification in helping teams to resolve this apparent paradox. The theory and results presented here therefore offer a hopeful agenda for future research on the costs and benefits of expertise diversity in multidisciplinary teams and for future practical applications designed to realize the potential benefits of team diversity.

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